

Research paper

Random forest with feature selection and K-fold cross validation for predicting the electrical and thermal efficiencies of air based photovoltaic-thermal systems

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ABSTRACT

The recent global energy transition policies emphasize the utilization of renewable sources as a primary energy solution. Solar energy systems, in particular, offer significant advantages in terms of cost-effectiveness and environmental friendliness. This work focuses on the modeling of a photovoltaic-thermal (PV/T) air-based system using machine learning (ML) techniques. The primary contribution lies in estimating the electrical and thermal efficiencies of the PV/T system through the application of the random forest (RF) technique and K-fold cross-validation. The results obtained are notably impressive when compared to the most commonly used techniques reported in the literature: Artificial Neural Networks (ANN), Support Vector Machine (SVM), and linear regression (LR). The assessment demonstrates that Random Forest outperforms these techniques in predicting both Electrical efficiency and Thermal efficiency with R^2 values of 99.9996 % and 81.2 % respectively, and MAE of 0.0034 and 2.64 respectively. Furthermore, selecting the most important features for electrical efficiency not only reduced the processing time but also improved the performance. The Random Forest model exhibited robust stability and demonstrated strong generalization capabilities, as evidenced by its consistent and promising electrical efficiency performance across a 10-fold cross-validation assessment. The proposed method could be a key solution to accurately model the electrical and thermal performances and serve as an alternative to traditional physical modeling approaches.

1. Introduction

During the last decades, the demand for energy has increased dramatically. According to the International Energy Agency (IEA), the demand for electricity is projected to reach 30,000 TWh by 2025 (IEA, 2023). In pursuit of achieving carbon neutrality objectives, renewable energy sources offer a green solution with various advantages and energy perspectives. Solar energy, in particular, offers a range of mature technologies that have been widely adopted. One of these highly effective technologies is Photovoltaic-Thermal (PV/T) energy systems. The main advantage of PV/T systems is the simultaneous generation of electricity and heat using a single system (Chaibi et al., 2021).

A PV/T system includes a cooling device specifically intended to disperse the heat generated by photovoltaic cells when they are exposed to sunshine (Abdelrazik et al., 2018). The purpose of this device is to decrease the operational temperature of the PV cells, hence improving

their efficiency in generating electricity while also producing thermal heat (Siecker et al., 2017). Generally, these devices can be classified into two categories: air-based PV/T systems and water-based systems, based on the type of heat transfer fluid (HTF) employed to extract the thermal heat (Siecker et al., 2017; Vaishak and Bhale, 2019). When it comes to simplicity of use and maintenance, air is considered more suitable than water as a cooling medium, and it also has lower energy consumption compared to water cooling (Chaibi et al., 2021). For hybrid air-based PV/T systems, the installation process is nearly identical to standard PV systems, with the key difference being the inclusion of an air layer between the panel and the absorber (Tiwari et al., 2018). There's the option to either rely on natural air circulation or implement forced ventilation to enhance thermal extraction and cooling of the photovoltaic cells. Numerous studies, both theoretical and experimental, have focused on characterizing thermal exchanges in these air-based PV/T panels, predicting their performance, and seeking optimal

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configurations to maximize thermal and electrical efficiencies (Chaibi et al., 2021; Tonui and Tripanagnostopoulos, 2007; Sathe and Dhoble, 2017). Some researchers have opted to use an electrical analogy method to analyze these collectors. This approach helps to simplify the complex interactions between the airflow and the many layers of the PV/T air system. Other scientists have employed a comparable methodology to that utilized in traditional air collectors (Solanki et al., 2009). Several authors have developed dynamic physical models to characterize the energy performance of the PV/T air panel. These models are based on energy conservation equations for the various layers comprising the system (Chaibi et al., 2021; Joshi et al., 2009; Shan et al., 2014). Energy calculations depend on the thermal interaction among the airflow, the components of the collector, and external factors such as wind speed, humidity, sunlight, and outdoor temperature. These physical models enable tracking the evolution of both thermal and electrical efficiencies, as well as the air temperature at the panel outlet, in response to variations in solar irradiance and ambient temperature. It is important to mention that most of these models produce inaccuracies when compared to experimental results. These inaccuracies are often related to the system's complexity, its thermal inertia, various thermal losses of the collector, and the different assumptions employed. Consequently, many academics are exploring new approaches that use artificial intelligence algorithms to address the limitations caused by these physical models and optimize the energy efficiency of these collectors (Shan et al., 2014; Zamen et al., 2019; Kalani et al., 2017).

Recent advancements in PV/T systems have concentrated the efforts on improving their efficiency and modeling their performance accurately (Zamen et al., 2019; Al-Waeli et al., 2018; Kalani et al., 2017; Al-Waeli et al., 2019; Shaban et al., 2024; Deka et al., 2024; Hai and Zhou, 2023; Deka et al., 2023; Ali et al., 2023). In the study presented in (Al-Waeli et al., 2018), Al-Waeli et al. evaluated a PV/T system with three cooling approaches: water, Phase Change Material (PCM) with water, and nano-PCM with nanofluid. It was found that the incorporation of nano-PCM and nanofluid enhanced the electrical efficiency from 8.07 % to 13.32 %, marking a significant 5.25 % increase compared to conventional PV systems. To model the system's performance, three Artificial Neural Network (ANN) methods were employed: MLP, SOFM, and SVM. The MLP model exhibited the lowest mean squared error (MSE) for both current prediction (0.01226) and efficiency prediction (0.03758), suggesting its superiority in modeling accuracy over the SOFM and SVM models, which had higher MSE values.

Another study investigated a Building Integrated Photovoltaic Thermal (BIPVT) system with a spiral flow absorber, designed to generate both electricity and hot water. Through energy and exergy analyses, the system demonstrated PV/T energy efficiencies of 55–62 % and exergy efficiencies of 12–14 %. Moreover, the system's primary-energy saving efficiency ranged from 73 % to 81 %, highlighting its potential in warm-humid climates. However, the performance of this BIPVT system is dependent on weather conditions, and there is a need for further optimization of the thermal absorber's design and materials. In a different approach (Mojumder et al., 2016). The study presented a support vector machine (SVM) model for predicting the electrical and thermal performance of two types of PV/T systems. The SVM-Wavelet model, in particular, provided the most accurate predictions, achieving an R^2 value of 0.5887 for electrical efficiency prediction in Design A and 0.9869 for Design B. For overall thermal efficiency, the R^2 values were 0.9805 for Design A and 0.8871 for Design B. Despite its high predictive accuracy, the SVM-Wavelet model's accuracy depends on the quality of experimental data, and further validation is required for different PV/T system designs and operating conditions. Researchers in (Mojumder et al., 2017) used soft computing to estimate electrical and thermal efficiency in a PV/T system. Experimental data was collected out of a two-air type PV/T designs that were practiced for a cooling fin design in the collector. Data was leveraged to train three models, Artificial Neural Network (ANN), Genetic Programming (GP) and Extreme Learning Machine (ELM). The later stands out as the best performer with

an R^2 of 98.75 % and 98.61 % respectively for electrical efficiency and thermal efficiency. Another study that uses Artificial Neural Network is (Al-Alawi et al., 2007). This paper presents an ANN-based controller for optimizing a hybrid renewable energy system with photovoltaic modules, a diesel generator, a battery bank, and a desalination unit. The controller improves efficiency, reduces fuel use and emissions, and accurately predicts power usage and the ON/OFF status of the diesel generator. It achieved a prediction accuracy with an R^2 value of 0.979 from 186 test cases. In (Argiriou et al., 2004), authors present an ANN-based controller for building hydronic heating plants that forecasts ambient temperature, solar irradiance, and optimizes water supply temperature. Tested on an office building and via TRNSYS simulations, it achieved about 15 % energy savings compared to a conventional controller in North European weather conditions. The study in (Ravaee et al., 2012) leveraged an ANN model for a Photovoltaic Thermal (PV/T) collector, using inputs like ambient and cell temperature, fluid temperature and velocity, solar identity, and time, with outputs of thermal and electrical efficiency. The optimal model, featuring one hidden layer with 10 neurons, was trained using data measured in Zahedan conditions and the Levenberg-Marquardt backpropagation algorithm. The results demonstrate that this ANN approach accurately predicts PV/T performance, requiring only limited experimental data, with an R^2 of 99.76 % and an RMSE of 0.39 for electrical efficiency, while an R^2 of 98.22 % and an RMSE of 0.134 for thermal efficiency. In (Kalani et al., 2017), a novel approach was introduced, the authors modeled a photovoltaic thermal nanofluid-based collector (PV/T/N) system using ANNs and particle swarm optimization (PSO). Employing three ANNs, including radial-basis function ANN (RBFANN), multi-layer perceptron ANN (MLPANN), and adaptive neuro-fuzzy inference system (ANFIS), the study demonstrated that these models could accurately predict fluid outlet temperature and electrical efficiency. The ANFIS model showed higher accuracy in predicting fluid outlet temperature with an RMSE of 0.3638 % and an R^2 of 0.9934, while the RBFANN model was more accurate for electrical efficiency with an RMSE of 0.2562 % and an R^2 of 0.9906. Further, mathematical and neural network models were developed to predict the electrical performance of a PV/T system utilizing nano-PCM and nanofluid cooling (Al-Waeli et al., 2019). The best neural prediction models achieved a commendable R^2 score of 0.81 and an MSE of 0.0361. This study underscores the potential of mathematical and neural network models in accurately predicting the future performance of PV/T systems, providing valuable tools for designers and manufacturers. Lastly, the study presented in (Zamen et al., 2019); employed artificial intelligence algorithms, specifically MLP-ANN, ANFIS, and LSSVM, to estimate the thermal efficiency of a PV/T system. The models were trained and validated using experimental data, with the ANN model showing the best performance with an MSE of 0.009 and an R^2 of 1.00. Sensitivity analysis indicated that inlet temperature had the most significant impact on thermal efficiency, with a relevancy factor of 0.6.

These studies collectively highlight the advancements in PV/T system design, cooling techniques, and performance modeling. The incorporation of nano-PCM, nanofluids, and advanced modeling techniques such as ANNs and SVMs has shown promise in enhancing the efficiency and predictive accuracy of PV/T systems. However, challenges such as system scalability, long-term stability of nanofluids, and the need for further optimization and validation of models remain areas for future research. In this context, the particularity of this paper lies in the use of Random Forest with k-fold cross-validation to predict the electrical and thermal efficiencies of a PV/T air-based system, which aims at enhancing our previous work (Chaibi et al., 2021), by improving the predictive accuracy and designing a more credible model. In fact, the choice of Random Forest is supported by a comparative study to other methods from the literature, both with and without k-fold cross-validation. The assessment of these techniques is carried out using validated data from previous published works, demonstrating that Random Forest with k-fold cross-validation outperforms other methods reported in the literature.

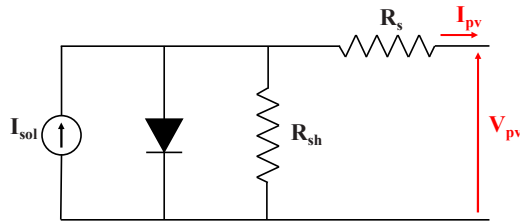


Fig. 1. The electrical model of PV cell/module.

2. Methodology and data

2.1. Electrical and thermal model of the PV/T system

In this section, we assess the validated models adopted in the prediction of the electrical and thermal performance. The electrical aspect is modeled using one of the most commonly employed equivalent circuit models, specifically the single-diode configuration presented in Fig. 1. The mathematical modeling to determine the single-diode electrical parameters is based on experimental measurements and the mathematical formulations provided in the manufacturer's datasheet equations. Subsequently, the calculated electrical parameters from the model are employed directly to compute the electrical performance for various input values of temperature and solar irradiance.

The prediction of PV power involves understanding the evolution of both the PV current (I_{pv}) and voltage (V_{pv}) in response to climatic variations. Therefore, applying Kirchhoff's law to the circuit diagram in Fig. 1 yields the expression for the output PV current as follows:

$$I_{pv} = I_{sol} - I_s \left\{ \exp \left[\frac{q}{AkT_{cell}N_{cell}} (V_{pv} + I_{pv}R_s) - 1 \right] \right\} - \frac{V_{pv} + R_s I_{pv}}{R_{sh}} \quad (1)$$

where, k and q represent the Boltzmann constant and the electron charge, respectively, while I_{sol} represents the photo-generated current. This current is influenced by climatic variations and can be computed using Eq. (2):

$$I_{sol} = [I_{sc} + K_i(T_{cell} - 298.15)] \frac{G}{1000} \quad (2)$$

Where, I_{sc} is the short-circuit current of the PV module at standard test conditions (STC), and K_i is the temperature coefficient associated with this current. To determine the values of I_{pv} and V_{pv} a set of two equations is required. In addition to Eq. (1), another equation is derived using operation at the maximum power point (MPP) as follows (De Soto et al.,

2006):

$$\frac{\partial P_{pv}}{\partial V_{pv}} = I_{pv} + V_{pv} \frac{\partial I_{pv}}{\partial V_{pv}} = 0 \quad (3)$$

Thus, Eq. (3) can be expressed as:

$$I_{pv} = (V_{pv} - R_s I_{pv}) \left\{ \frac{q}{AkT_{cell}N_c} I_s \exp \left[\frac{q}{AkT_{cell}N_c} (V_{pv} + R_s I_{pv}) \right] + 1/R_{sh} \right\} \quad (4)$$

Furthermore, the computed value of the reverse saturation current, I_r , can be used to calculate the saturation current, I_s , for a different range of temperatures:

$$I_s = I_r \left(\frac{T_{cell}}{298.15} \right)^3 \exp \left(\frac{qE_{G0}}{Ak} \left[\frac{1}{T_{cell}} - \frac{1}{298.15} \right] \right) \quad (5)$$

where, E_{G0} represents is the silicon band-gap (=1.12 eV).

The need to assess the electrical performance of the PV panel and its evolution in the proposed air collector PV/T system necessitates the use of Eq. (6). The $\eta_{electrical}$ value depends on the variations in the generated power, P_{pv} and solar irradiance, G . Additionally, S_m represents the PV module area.

$$\eta_{electrical} = \frac{P_{pv}}{S_m G} \quad (6)$$

The thermal model adopted in this study relies on the main assumptions:

- Transient regime
- Unidirectional airflow refers to the movement of air in a single direction.
- The temperature of the glass, cell, Tedlar, and air depends solely on the longitudinal dimension in the flow direction.

This model has been thoroughly detailed in a previous article (Chaibi et al., 2021; Chaibi et al., 2021), and there is no need to represent it in the scope of this work. It enables the determination of the temporal evolution of the temperature of the air, glass, cell, and Tedlar in the direction of the flow. It also allows for the calculation of the thermal efficiency of the PV/T system, as described by the following equations:

$$\eta_{thermal} = \frac{C_f \dot{m}_f (\bar{T}_f - T_{f,in})}{GS_m} \quad (7)$$

where, $\bar{T}_f$ is the average temperature of the fluid, $\dot{m}_f$ is the mass flow rate

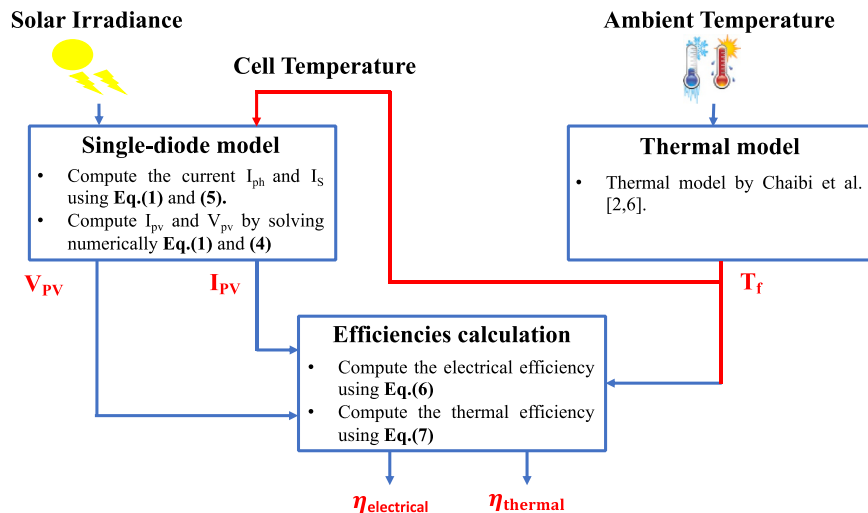


Fig. 2. The flowchart used to compute the electrical and thermal efficiencies.

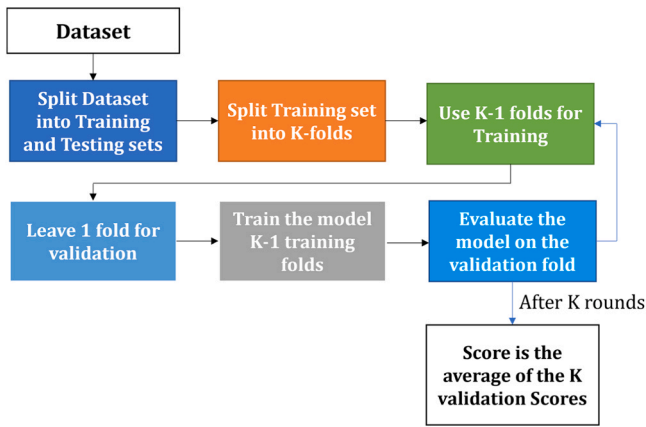


Fig. 3. K-Fold Cross-Validation Process.

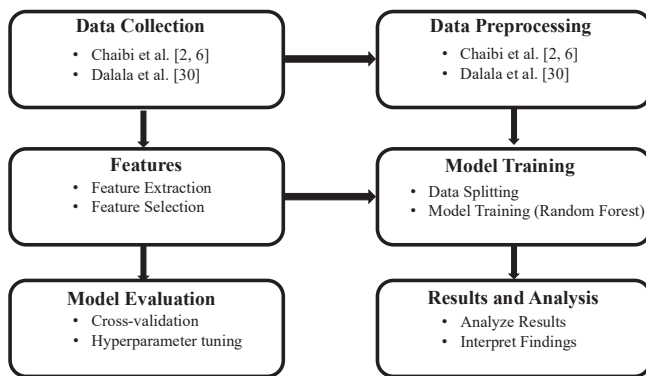


Fig. 4. Flowchart of the process from data collection to analysis.

Table 1

Best Hyper-parameters configuration for each tested ML model.

ML model	Hyper-parameters configuration
LR	- Ridge regularization term α : 0.5
SVM	- Kernel: Radial Basis Function - C: 1 (Regularization parameter) - γ : 1/n (Kernel coefficient for 'rbf')
ANN	- Number of hidden layers: 2 - Number of units in each layer: [128–64–1] - Activations in each layer: [relu, relu, linear] - Optimizer: Adam - Loss: Mean squared error - Epochs: 150 - Batch size: 32
RF	- Number of estimators: 50 - Criterion: Mean squared error - min_samples_split: 2 - min_samples_leaf: 1 - max_depth: Nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples - max_features: Number of features

of air and the electrical energy conversion factor C_f ranges from 0.35 to 0.40 and is generally used for PV/T systems (Evans, 1981).

The adopted equations for the electrical and thermal performances calculations are explained in the flowchart in Fig. 2.

Please note that the adopted models to compute both efficiencies are based on validated models from the literature, more information are presented in (Chaibi et al., 2020) and (Chaibi et al., 2021).

2.2. Machine learning techniques for performances prediction

In this section, we introduce machine learning techniques, specifically Ridge Multiple linear Regression (LR), Artificial Neural Networks (ANN), Support Vector Machines (SVM) with RBF Kernel, and Random Forest (RF). These regression approaches are applied to forecast both the electrical and thermal efficiency of a photovoltaic-thermal air collector system.

2.2.1. Multiple linear regression

In multiple linear regression, the objective is to find the best-fitting linear equation that describes the relationship between the dependent variable $\hat{y}$ and two or more independent variables ($x_1, x_2, \dots, x_n$). The general form of the multiple linear regression equation is (Krzywinski and Altman, 2015):

$$\hat{y} = w_0 + w_1 \cdot x_1 + w_2 \cdot x_2 + \dots + w_n \cdot x_n \quad (8)$$

where, w_0 is the $\hat{y}$ -intercept, meaning the value of $\hat{y}$ when all x_i values are zero.

$w_0, w_1, \dots, w_n$ are the coefficients that represent the change in $\hat{y}$ for a one-unit change in the corresponding x_i variable, assuming all other variables are held constant.

Ridge regression is a type of linear regression that includes a regularization term to prevent overfitting in the presence of multicollinearity (high correlation between independent variables). The ridge regression equation is an extension of the multiple linear regression equation with an additional regularization term (Venkatesh et al., 2023):

$$\hat{y} = w_0 + \sum_{i=1}^n w_i \cdot x_i + \alpha \cdot \sum_{i=1}^n w_i^2 \quad (9)$$

where, α is the regularization parameter. The term $(\alpha \cdot \sum_{i=1}^n w_i^2)$ is the regularization term. It penalizes large coefficients by adding the squared values of the coefficients multiplied by the regularization parameter α to the ordinary least squares cost function. The purpose of this regularization term is to shrink the coefficients towards zero, especially when there is multicollinearity among the independent variables.

Ridge regression helps to stabilize the model and prevent it from becoming too sensitive to the specific values of the training data. The value of the regularization parameter α is a tuning parameter that controls the strength of the regularization. Finding an optimal value for α is crucial, and this is often done through techniques like cross-validation.

The loss function in ridge regression consists of two components: the traditional least squares error term and the regularization term is defined by (Venkatesh et al., 2023):

$$Loss = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \cdot \sum_{j=1}^p w_j^2 \quad (10)$$

where, y_i is the observed value, $\hat{y}_i$ is the predicted value, w_i are the weights, and α controls the strength of regularization.

2.2.2. Support Vector Machine (SVM) with RBF kernel

The Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel is a highly regarded learning approach widely employed to effectively tackle overfitting challenges. In this SVM formulation, the outputs y_i undergo a transformation through a regression function φ acting on the inputs x_i as specified by the equation (Ding et al., 2021):

$$y_i = \omega \cdot \varphi(x_i) + b, \quad i = 1 \dots N \quad (11)$$

Here, ω represents the weight vector, b is the bias, and N is the size of the training dataset. To address the optimization problem and minimize the cost function, the Lagrange function is introduced.

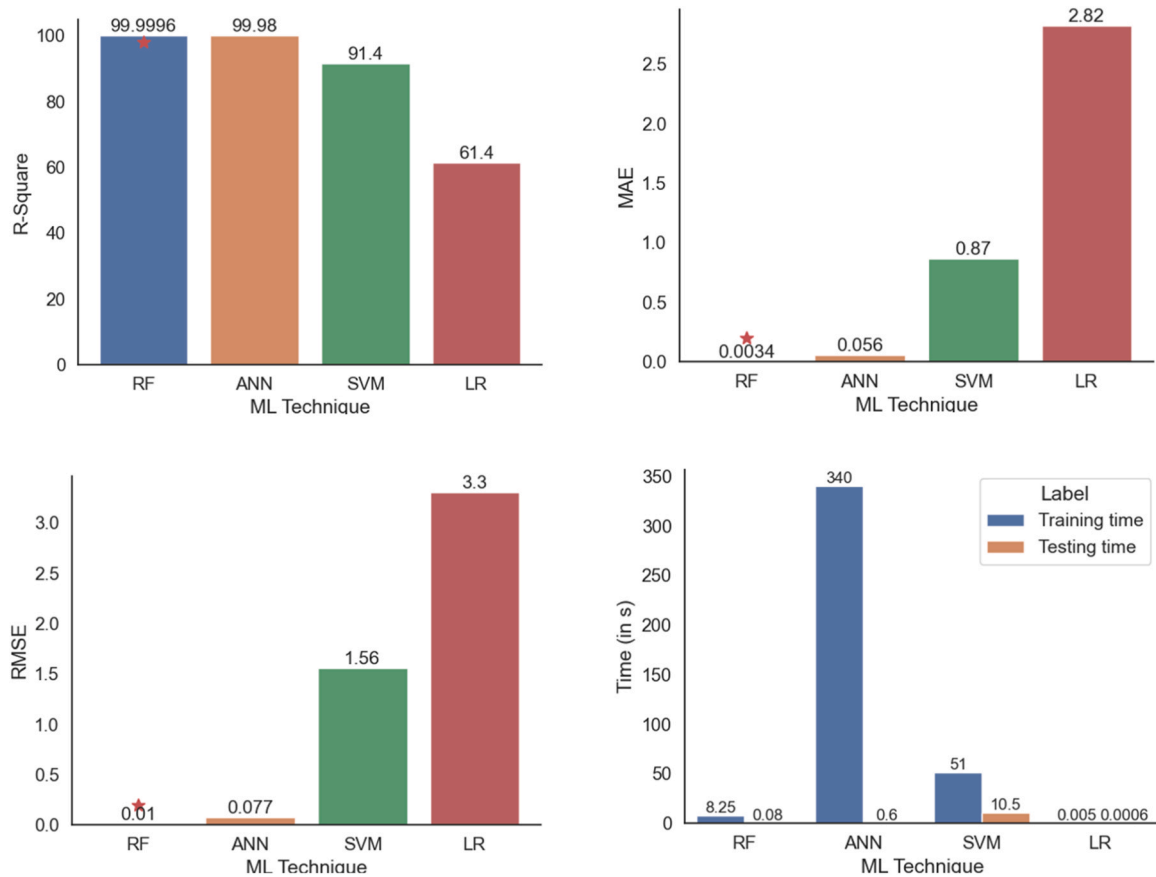


Fig. 5. Performance Evaluation of the electrical efficiency of the PV/T system for RF, ANN, SVM and LR, without K-Fold Cross-Validation.

$$L(\omega, b, e, \alpha) = J(\omega, e) - \sum_{i=1}^n \alpha_i [y_i(\omega \cdot \varphi(x_i) + b) - 1 + e_i], \quad i = 1 \dots N \quad (12)$$

The cost function:

$$J(\omega, e) = \frac{1}{2} \omega^T \omega + \frac{\gamma}{2} \sum_{i=1}^n e_i^2, \quad i = 1 \dots N \quad (13)$$

The e_i and α_i are unknown variables, and γ is the regularization factor. Mercer's theorem introduces a kernel function K and asserts that (Ding et al., 2021):

$$\varphi^T(x_i) \varphi(x_j) = K(x_i, x_j), \quad i, j = 1 \dots N \quad (14)$$

The Radial Basis Function (RBF) is assumed as the kernel function, defined by (Ding et al., 2021):

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right), \quad i, j = 1 \dots N \quad (15)$$

The σ represents the tuning parameter. As a result, an approximation of y is expressed as (Ding et al., 2021):

$$y(x) = \sum_{i=1}^N \alpha_i K(x, x_i) + b \quad (16)$$

2.2.3. Artificial Neural Network (ANN)

ANNs are computational models inspired by the structure and functioning of the human brain. ANNs consist of interconnected nodes called artificial neurons, organized in layers: an input layer, one or more hidden layers, and an output layer. The fundamental unit of an ANN is the artificial neuron. The activation of a neuron a_i in a layer is computed by applying an activation function f to the weighted sum of its inputs (Montesinos López et al., 2022):

$$a_i = f\left(\sum_{j=1}^n w_{ij} \cdot x_j + b_i\right) \quad (17)$$

where, w_{ij} is the weight between neuron i and output j . The variable x_i is the input from neuron j in the previous layer, and b_i is the bias neuron i .

During the training, input data is fed into the input layer, and computations propagate through the hidden layers to produce an output. The feedforward process can be expressed mathematically as (Montesinos López et al., 2022):

$$a^{(l)} = f(W^{(l)} \cdot a^{(l-1)} + b^l) \quad (18)$$

where, $a^{(l)}$ is the activation vector of layer l , $W^{(l)}$ represents the weight matrix for layer l , and b^l is the bias vector for layer l .

ANNs learn by adjusting the weights and biases on the error between the predicted output and actual target. The process of updating weights to minimize this error is typically achieved through backpropagation and optimization algorithms like gradient descent (Montesinos López et al., 2022).

$$w_{ij}^{(l)} \leftarrow w_{ij}^{(l)} - \eta \frac{\partial L}{\partial w_{ij}^{(l)}} \quad (19)$$

where, η is the learning rate, and L represents loss function.

2.2.4. Random Forest (RF)

Random Forest is an ensemble of decision trees, and the process begins with the construction of individual trees. Let N denote the number of trees in the forest, M be the total number of features in the dataset D , and m be the number of features considered at each split. For each tree I , a bootstrap sample D_i is drawn with replacement from the

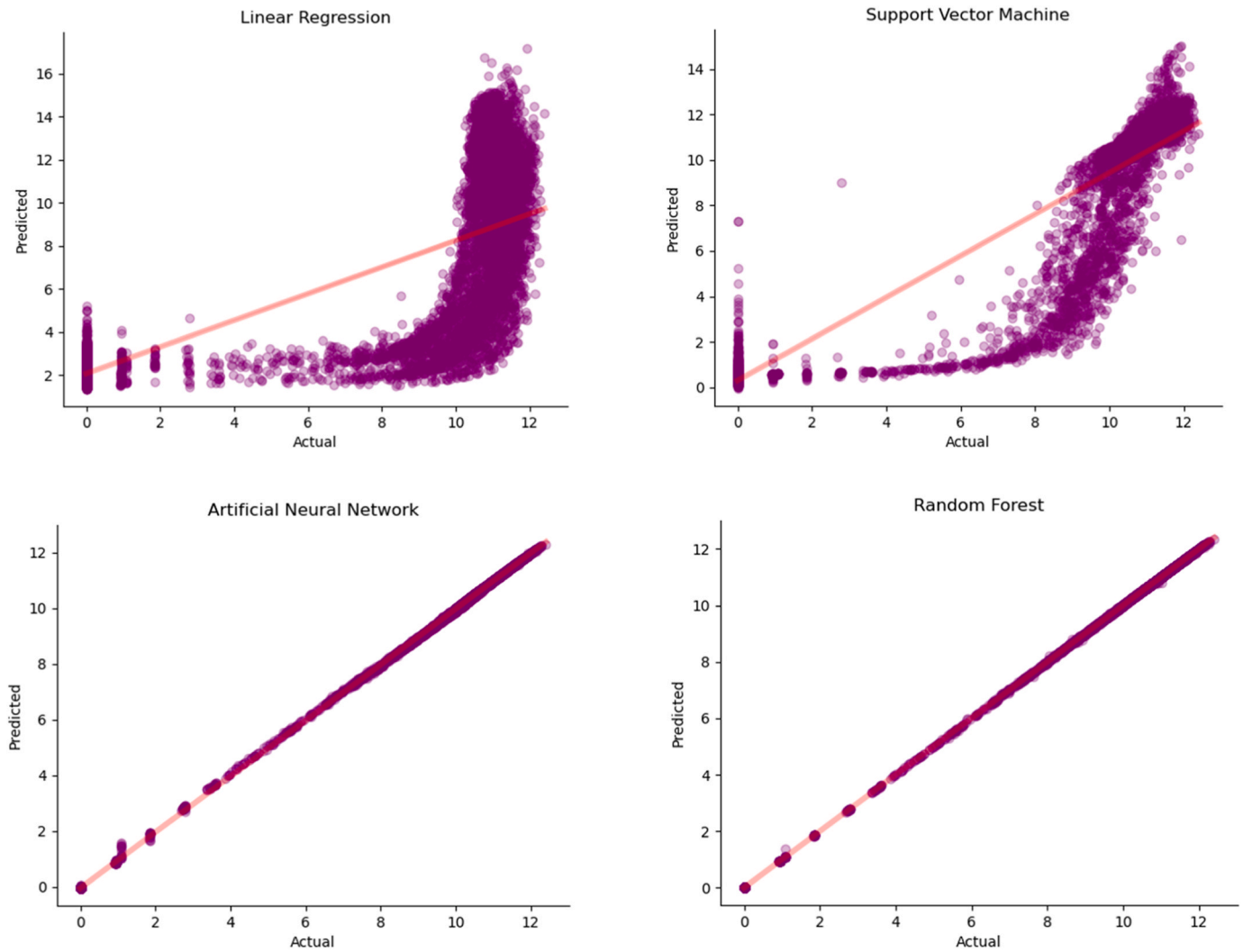


Fig. 6. Actual VS Predicted Electrical Efficiency of the PV/T system for RF, ANN, SVM and LR.

original dataset D , and a random subset of features m is chosen at each split (Borup et al., 2023).

At each terminal node of a tree, a response R_i is assigned based on the majority class for classification or the average for regression. The prediction of the i -th tree, denoted $T_i(X)$, is determined by traversing the tree:

$$T_i(X) = R_i \quad (20)$$

The algorithm aims to minimize the variance while maintaining low bias. For classification, it often employs the Gini impurity, and for regression, it minimizes the mean squared error presented in the equations below (Borup et al., 2023):

$$Gini(D) = \sum_{k=1}^K p_k(1 - p_k) \quad (21)$$

$$MSE(D) = \frac{1}{|D|} \sum_{i \in D} (y_i - \bar{y}_d)^2 \quad (22)$$

where, K is the number of classes, p_k is the proportion of class k instances in node D , $|D|$ is the number of instances in node D , y_i is the target value for instance i , and $\bar{y}_d$ is the mean target value in node D .

2.3. Normalization of data

As a feature scaling technique, Min-Max normalization was employed to rescale the input data in the range (0,1). Because the features in the data have different value ranges, we employed feature scaling to normalize all features to the same range. This ensures that our model doesn't exhibit bias towards specific features solely due to their larger values. We scaled the features using the Min-Max normalization approach according to the following equation (Han et al., 2012):

$$x_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (23)$$

where, x_i represents the numerical feature value of the i^{th} sample, while $x_{\min}$ and $x_{\max}$ denote the minimum and maximum values of each numeric feature, respectively.

2.4. K-fold cross validation

In ML, generalization refers to the ability of a model to adapt to new unseen data, thus producing accurate predictions. Generally, if the new unseen data fed into the model, has similar characteristics with the training data, the model performs well, and vice-versa (Alapatt et al., 2020). Cross-validation serves as a valuable tool for assessing machine learning models. It consists of training of multiple models on various subsets of the input data. Cross-validation is employed to identify

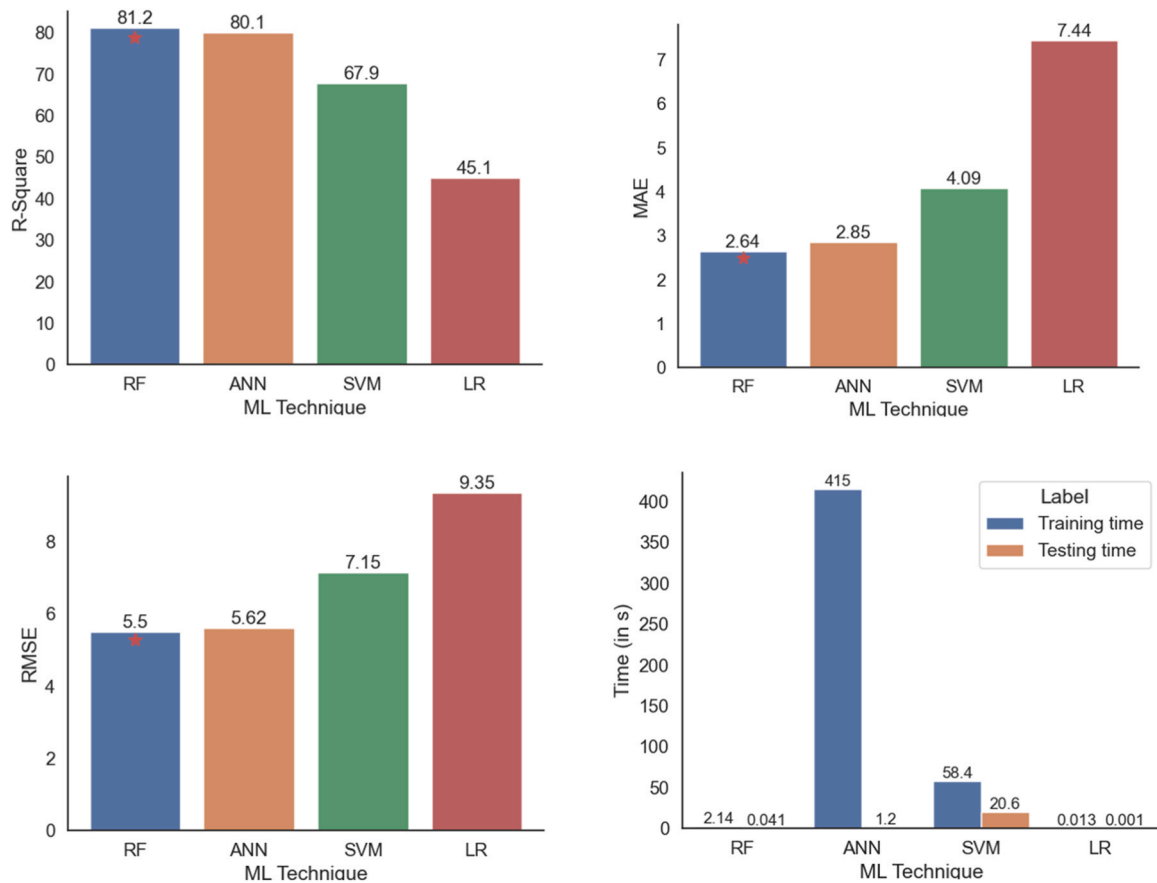


Fig. 7. Performance Evaluation of the thermal efficiency of the PV/T system for RF, ANN, SVM and LR, without K-Fold Cross-Validation.

overfitting in a model, which infers that the model is struggling to effectively generalize patterns and similarities in new unseen input data. Cross-validation can be performed based on a plethora of methods, among which K-Fold is a popular one (Wong, 2015). Basically, K-Fold consists of the following steps (See Fig. 3):

- 1) Split dataset into Training and Testing subsets
- 2) Choose K
- 3) Split the Training dataset into K subsets (folds)
- 4) In the 1st round, set K-1 Folds for Training, and a different Fold for Validation.
- 5) Train the model on the K-1 Folds, validate on validation Fold, and save the validation score
- 6) Perform K-1 additional rounds by repeating steps 4–5, but with changing the validation fold at each round
- 7) After validating the model on all the K folds, the final score is obtained by averaging the validation scores obtained in each round.

The following figure summarizes the K-Fold Cross-Validation process:

2.5. Performance metrics

Evaluation of the performance is based on:

2.5.1. R-square

R^2 is a statistical measure indicating the proportion of the variance in the dependent variable explained by the independent variables in a regression model (Krzywinski and Altman, 2015). In other words, it represents the goodness of fit of the model

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (24)$$

where y_i is the observed value of the dependent variable for sample i , $\hat{y}_i$ is the predicted value of the dependent variable for sample i based on the regression model, $\bar{y}$ is the mean of the dependent variable, and n is the number of observations.

2.5.2. Mean Absolute Error (MAE)

MAE represents the average absolute difference between predicted values and observed values of the dependent variable. MAE does not emphasize large errors over smaller ones, treating all errors equally in its computation (Karunasingha, 2022):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (25)$$

where, y_i is the observed value of the dependent variable for sample i , $\hat{y}_i$ is the predicted value of the dependent variable for sample i based on the regression model, and n is the number of observations.

2.5.3. Root Mean Square Error (RMSE)

RMSE is another metric for evaluating a regression model. It penalizes large errors more heavily than MAE, because of the squared difference. Hence, RMSE is more sensitive to outliers and may be more influenced by large errors in the predictions (Karunasingha, 2022).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (26)$$

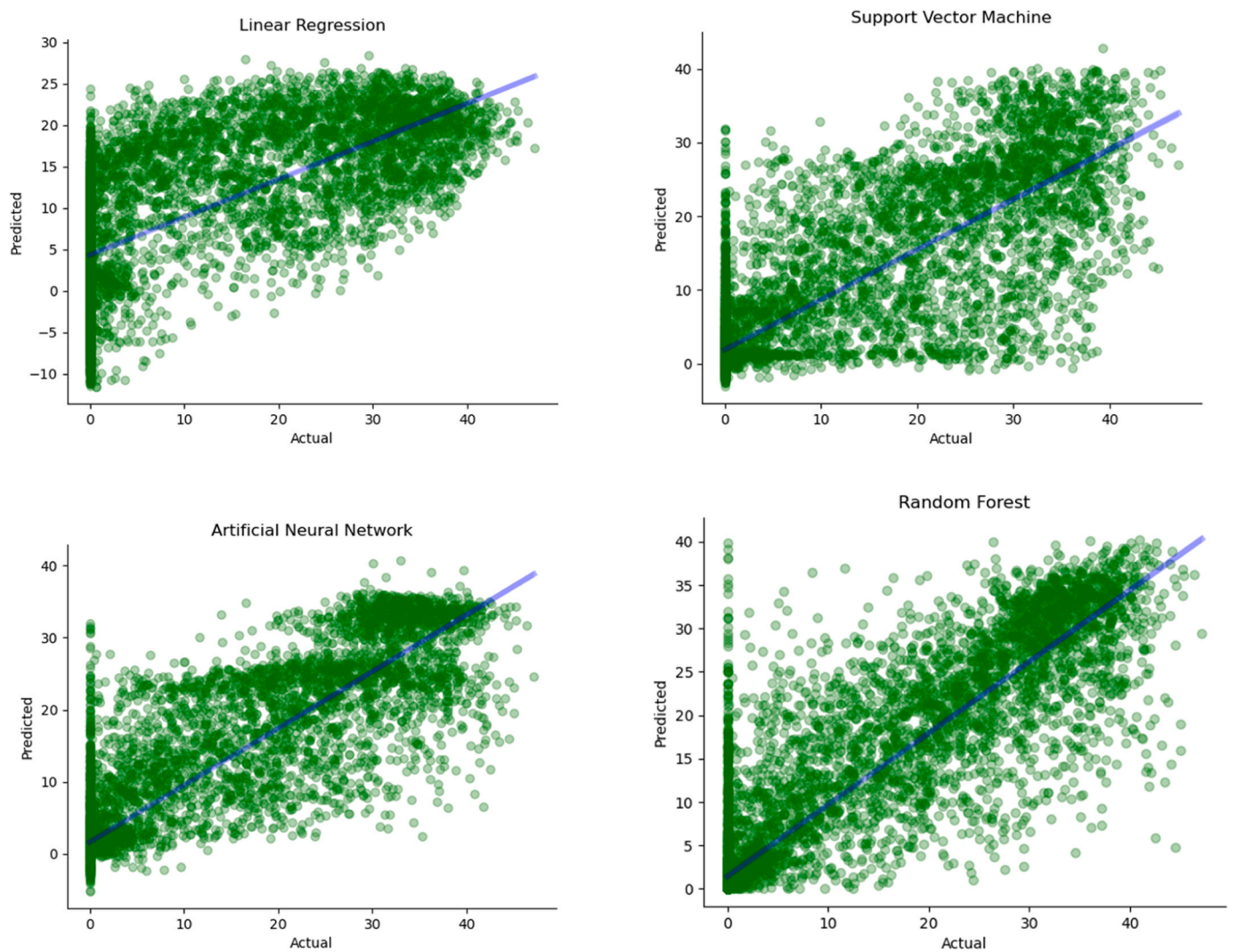


Fig. 8. Actual VS Predicted thermal efficiency of the PV/T system for RF, ANN, SVM and LR.

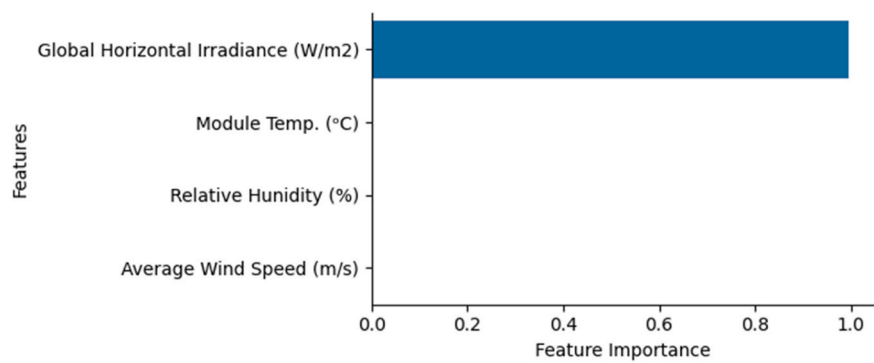


Fig. 9. Feature Importance in predicting Electrical Efficiency of the PV/T system with RF model.

where y_i is the observed value of the dependent variable for sample i , $\hat{y}_i$ is the predicted value of the dependent variable for sample i based on the regression model, and n is the number of observations

2.5.4. Training and testing time

Training time is the period during which a ML model is fed with data to learn the patterns and relationships within the data. Testing time is when a trained ML model is used to make predictions on new unseen

data

Basically, the process consists of first, training the four aforementioned machine learning models to estimate both electrical and thermal efficiencies, without using cross-validation. We then select the best model in term of performance results and train it based on 10-Fold cross-validation, to assess its capacity of generalization. We further analyze the selected model to improve its outcome with respect to the performance metrics, by selecting only the most important features among the

Table 2
Electrical Efficiency Prediction results of the PV/T system with RF model before and after Feature Selection.

Performance metric	Before Feature Selection	After Feature Selection
MAE	0.034	0.029
RMSE	0.01	0.0089
R ² (%)	99.9996	99.99971
Training Time (s)	8.25	5.38
Testing Time (s)	0.08	0.07

feature set. Finally, we compare our result with findings from the literature.

3. Results and discussions

In this section, we compare the results obtained from LR, SVM, ANN, and RF. We also report results from (Chaibi et al., 2021) and compare our findings to those of a similar research scope from the literature (Chaibi et al., 2021).

3.1. Data description

The study utilizes real measurements of climatic data Dalala et.al (Dalala et al., 2020), including solar irradiance, module temperature, humidity, and wind speed, recorded at a weather station located in the Jordan Valley (Hashemite Kingdom of Jordan) for the year 2017. The data, collected every 30 minutes resulting in a total of 12,238 samples, ensure a comprehensive and detailed dataset. The quality of the raw data is maintained through consistent monitoring and recording, adhering to standard meteorological practices. Solar irradiance data ranges from 0 to 1093.06 W/m² with a mean value of 229.52 W/m², while module temperature varies from 4.72°C to 51.82°C with a mean of 24.13°C, showcasing high accuracy with a Mean Absolute Error (MAE) of 1°C. Humidity and wind speed data provide essential context for understanding environmental conditions, with respective ranges of 11.33–99.81 % and up to 19.4 m/s. Pearson correlation coefficients indicate significant relationships between irradiance and efficiencies (electrical: 0.78, thermal: 0.53), validating irradiance as a crucial factor. The robust analysis and detailed data collection underpin the reliability of the input data for the analytical models, effectively minimizing uncertainties and ensuring the validity of PV-T parameter estimations and subsequent machine learning model inputs. The Flowchart in Fig. 4 illustrates the entire process from data collection to analysis has been included in data description section.

3.2. Performance results: without K-fold cross-validation

Basically, the process consists of first, training and fine-tuning the four aforementioned machine learning models to estimate both electrical and thermal efficiencies, based on the four features, Solar

Irradiance, Module Temperature, Humidity, and Wind Speed, without using cross-validation. To achieve this, data is split into 80 % and 20 % for training set and testing set respectively. Data is then normalized according to Eq. (23). The LR model is trained with a ridge regularization term $\alpha = 0.5$. The SVM model is based on RBF kernel. The ANN model consists of a 4-features input layer, 2 hidden layers containing 128 units, and 64 units respectively, and an output layer for efficiency

Table 3
10-fold CV Performance results using RF (Electrical Efficiency).

Performance metric	Average result	Standard Deviation
MAE	0.0034	0.0011
RMSE	0.011	0.005
R ² (%)	99.9996	3.84×10^{-6}

Table 4
10-fold CV Performance results using RF (Thermal Efficiency).

Performance metric	Average result	Standard Deviation
MAE	3.3	0.67
RMSE	6.5	0.9
R ² (%)	63.3	0.14

Table 5
Performance comparison with the literature (Electrical Efficiency).

Model	MAE	RMSE	Pearson R (%)
SVM-Wavelet (Mojumder et al., 2016)	-	0.058	99.34
SVM-RBF (Mojumder et al., 2016)	-	0.067	99.31
SVM-FFA (Mojumder et al., 2016)	-	0.064	99.25
RBFANN (Kalani et al., 2017)	0.005	0.0025	99.52
MLPANN (Kalani et al., 2017)	0.0063	0.0036	96.76
ANFIS (Kalani et al., 2017)	0.0048	0.0027	99.47
MLP-SOFM (Al-Waeli et al., 2019)	-	0.03667	81.66
ANN (Chaibi et al., 2021)	0.0078	-	99.998
RF	0.0029	0.0089	99.9999
RF + 10-Fold CV	0.0034	0.011	99.9997

Table 6
Performance comparison with the literature (Thermal Efficiency).

Model	MAE	RMSE	Pearson R (%)
SVM-Wavelet (Mojumder et al., 2016)	-	0.3	94.2
SVM-RBF (Mojumder et al., 2016)	-	0.334	92.69
SVM-FFA (Mojumder et al., 2016)	-	0.333	92.72
ANN (Zamen et al., 2019)	-	0.093	100
GP (Mojumder et al., 2017)	-	0.347	92.16
ANN (Chaibi et al., 2021)	3.3607	-	86.84
RF	2.6	5.4	90
RF + 10-Fold CV (our work)	3.3	6.5	80

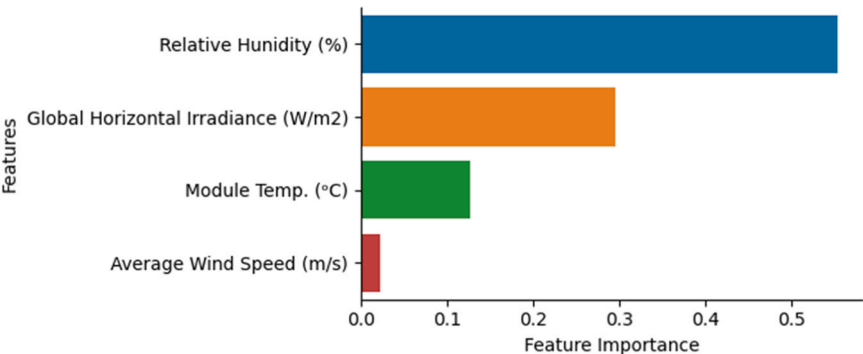


Fig. 10. Feature Importance in predicting Thermal Efficiency of the PV/T system with RF model.

prediction. The latter is trained with Adam optimizer, for 150 epochs, on training batches with 32 samples each, to minimize the mean square error loss function. Finally, the RF is trained by generating 50 decision tree estimators.

All the details about the hyper-parameters of each model are described in Table 1.

3.2.1. Electrical efficiency

Results for R^2 (in %), MAE and RMSE, are respectively reported in Fig. 5 as follows:

- **Explanatory Power:** RF and ANN significantly outperform LR and SVM, with RF having the highest R^2 value (99.9996 %), followed closely by ANN (99.98 %). LR and SVM have notably lower R^2 values, with SVM being better than LR in terms of explaining the variance in the data.
- **Prediction Accuracy:** RF and ANN again excel in this metric, with RF having the lowest MAE (0.0034) and ANN following closely with an MAE of 0.056. SVM performs reasonably well with an MAE of 0.87, while LR has the highest MAE of 2.82, indicating that RF and ANN provide the best accuracy in predictions. RMSE, similar to the MAE results, RF and ANN have the lowest RMSE values (0.01 and 0.077, respectively), signifying their superior predictive accuracy. SVM has a slightly higher RMSE of 1.56, while LR has the highest RMSE of 3.3.
- **Computational Efficiency:** In the context of training time, LR is the quickest, followed by RF. SVM and ANN require more time for training, with ANN being the slowest in this aspect. When it comes to testing time, LR is the fastest, followed by RF. SVM and ANN, while achieving high accuracy, demand more time during testing, but ANN necessitates additional time for training.

Considering both predictive accuracy and computational efficiency, RF emerges as an excellent choice, offering high accuracy and relatively fast training and testing times. LR provides the fastest training and testing times but sacrifices predictive accuracy. SVM, while accurate, demands significant time for both training and testing. ANN delivers a competitive accuracy but requires more time for both training and testing, posing a trade-off between accuracy and computational efficiency. The following plots in Fig. 6 show of the actual vs predicted electrical efficiencies, for LR, SVM, ANN, and RF.

3.2.2. Thermal efficiency

Results for R^2 (in %), MAE and RMSE, are respectively reported in Fig. 7 as follows:

- **Explanatory Power:** RF and ANN demonstrate superior explanatory power compared to LR and SVM. RF achieves the highest R^2 (81.2 %), indicating a strong ability to explain the variance in the dependent variable. ANN closely follows with an R^2 of 80.1 %. In contrast, LR and SVM have lower R^2 values (45.1 % for LR and 67.9 % for SVM), suggesting they capture less variance in the data.
- **Prediction Accuracy:** Both RF and ANN outperform LR and SVM in terms of prediction accuracy. RF and ANN exhibit lower Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), indicating smaller prediction errors. RF stands out for its relatively low MAE (2.64) and RMSE (5.5), emphasizing its accuracy.
- **Computational Efficiency:** LR is the most computationally efficient in terms of both training and testing time, with very short durations (0.013 seconds for training and 0.001 seconds for testing). SVM, while showing good accuracy, has the largest training and testing times (58.4 seconds and 20.6 seconds, respectively), making it computationally demanding. ANN has a longer training time (513 seconds) but compensates with a faster testing time (0.87 seconds). RF strikes a balance, providing moderate training time (2.14 seconds) and very fast testing time (0.04 seconds).

The following plots in Fig. 8 show the actual vs predicted electrical efficiencies, for LR, SVM, ANN, and RF.

3.2.3. Effect of feature selection

In the context of this work, RF was the most robust in handling complex relationships within the data, we selected it as the foundation for our predictive modeling endeavors, and our focus naturally turned towards maximizing its potential through strategic feature selection.

The initial Random Forest model incorporated four features, namely, "Global Horizontal Irradiance" and "Module Temperature", "Relative Humidity" and "Average Wind speed". Through a meticulous analysis of their impact on performance, we identified two pivotal variables for predicting the electrical efficiency — "Global Horizontal Irradiance" and "Module Temperature". Fig. 9 shows feature importance in predicting the electrical efficiency. These two features exhibited substantial influence on the model's outcomes, compelling us to delve deeper into their implications for further refinement.

With the overarching goal of fine-tuning the already robust Random Forest model for improved precision and efficiency, we conducted an in-depth evaluation of performance metrics. The subsequent sections provide a detailed comparison of the Random Forest model's performance before and after the strategic feature selection for electrical efficiency prediction (Table 2), highlighting the progress achieved through this targeted enhancement process.

The two critical features, "Global Horizontal Irradiance" and "Module Temperature," played a pivotal role in influencing the model's performance improvements.

Feature selection resulted in a notable decrease in MAE and RMSE, signifying an enhanced accuracy in predicting the dependent variable. The model's training time significantly decreased from 8.25 seconds to 5.38 seconds after feature selection, indicating a more efficient training process. Although we discarded two features from the feature set, we observed a slight increase in R-Square, which implies a marginal improvement in the model's ability to generalize to new, unseen data.

During this process, it became evident that two key variables "Global Horizontal Irradiance" and "Module Temperature", significantly influenced the model's outcomes for electrical efficiency prediction. However, it is crucial to note that when predicting thermal efficiency, we have determined, that all four features (including the aforementioned influential ones) contribute uniquely to the predictive accuracy of the model. The removal of any of these features led to a regression in the model's performance in predicting thermal efficiency. Fig. 10 shows feature importance levels for predicting thermal efficiency using RF model.

3.3. Performance results: using K-fold cross-validation

Having identified Random Forest (RF) as the most effective model among various algorithms in the context of this work, we have chosen it as the focal point for further analysis to ensure its generalizability. To do so, we have employed k-fold cross-validation as described in an earlier section, and leveraged the outcome of feature selection process. The subsequent analysis will present the results of 10-fold cross-validation, shedding light on the model's stability and generalization capacity.

3.3.1. Electrical efficiency

Results for R^2 (in %), MAE and RMSE, are respectively reported in Table 3 as follows:

- **Explanatory Power:** The R^2 value of 99.9996 % indicates an exceptionally high proportion of variance explained by the model. The very low standard deviation ($3.84 \cdot 10^{-6}$) suggests consistent performance across the folds. This implies that the model is robust in capturing the variability in the target variable and provides a high level of accuracy.

- **Prediction Accuracy:** The MAE value of 0.0034 with a standard deviation of 0.0011 suggests that, on average, the model's predictions deviate by only 0.0034 units from the actual values. The low standard deviation indicates consistency in performance across different folds. This is a positive sign, as it shows that the model is making accurate and reliable predictions. The RMSE value of 0.011 with a standard deviation of 0.005 indicates that, on average, the model's predictions have a root mean squared deviation of 0.011 units from the actual values. Similar to MAE, the low standard deviation suggests consistent performance. RMSE is more sensitive to larger errors, and this low value signifies that the model is effective in handling errors across different folds.

In summary, the consistently high R-square, low MAE, and low RMSE across the folds indicate that the Random Forest model performs exceptionally well in capturing the underlying patterns in the data. The negligible standard deviations for R-square, MAE, and RMSE suggest that the model's performance is stable and consistent, providing a high level of confidence in its ability to generalize to new, unseen data.

3.3.2. Thermal efficiency

Results for R^2 (in %), MAE and RMSE, are respectively reported in Table 4 as follows:

- **Explanatory Power:** The R^2 value (63.3 %), while indicating good explanatory power, also signals that there might be unaccounted-for factors contributing to the variance.
- **Prediction Accuracy:** The moderate MAE and RMSE values suggest that, on average, the model's predictions are reasonably close to the actual values. However, the standard deviations highlight that there are scenarios where the model's performance varies, indicating areas for potential improvement.

The results collectively indicate that the RF model provides a reasonably good fit to the data, explaining a substantial portion of the variability. However, the variability observed in the standard deviations across the metrics suggests that the model's performance is not entirely consistent across different subsets of the data.

3.4. Comparison with literature

In this section, we compare the performance and insights from our RF model with findings from the literature, and in particular, our previous work (Chaibi et al., 2021), as we strive to improve the predictive accuracy based on the same dataset. By aligning our results with established methodologies, our aim is to contribute to the ongoing body of knowledge and offer valuable insights.

Researchers in (Chaibi et al., 2021) use either MAE or RMSE, and the Pearson Correlation Coefficient (R) to evaluate predictive accuracy. Notably, in (Chaibi et al., 2021), the standout performer is an ANN model, representing the best outcome. As we compare results with later, we consider both the strengths of cross-validation in our model and the insights derived from MAE and the Pearson Correlation Coefficient (R) in the referenced article. Table 5 and Table 6, report Pearson Correlation Coefficient (R) (in %), MAE/RMSE for both electrical efficiency and thermal efficiency respectively.

For electrical efficiency prediction (See Table 5), both models exhibit strong relationship between predicted and observed values, with a slight advantage to the RF model. In terms of MAE, our RF model outperforms the ANN model in terms of accuracy, as evidenced by a low MAE (0.0034 vs 0.0078). For thermal efficiency in Table 6, both models perform competitively, with ANN showcasing stronger linear relationship between predicted and observed values, when comparing with our cross-validated results. Without using cross-validation, our model achieves an R^2 of 81.2 % (Fig. 7), an MAE of 2.64 (Fig. 7), and a calculated Pearson Correlation Coefficient (R) of 90 %, which outperforms the

ANN model. It is worth to note, that the absence of a validation approach in the all the models presented in Table 5 and Table 6 raises concerns about potential overfitting and generalizability to new unseen data.

Overall, for electrical efficiency, RF demonstrates superior accuracy compared to the ANN model (Chaibi et al., 2021). For thermal efficiency, ANN exhibits stronger linear relationship compared the cross validated RF model, but is outperformed by the later in case it is not cross validated. The inclusion of a 10-fold cross validation process in our RF model enhances its reliability and generalizability, providing a crucial advantage over the ANN model with no specified validation approach.

4. Conclusion

The physical modeling of a PV/T system is a complex task that requires mathematical formulation of both the electrical and thermal aspects. The main particularity of this study is to consider the use of K-fold cross-validation and Feature selection with random forest to predict the electrical and thermal efficiencies of a PV/T air-based system. This technique is compared to other commonly used methods from the literature (ANN, SVM, and LR) and demonstrates high performance. The main conclusions of this study are:

- The random forest with K-fold cross-validation outperforms other techniques from the literature in terms of prediction, as demonstrated by R-square, MAE, and RMSE.
- The dependency of electrical efficiencies is relative to irradiance and temperature variation, while for thermal efficiencies, humidity plays an important role in prediction.

In addressing the challenge of predicting thermal efficiency more accurately, future research efforts could explore other modeling strategies and data enhancement techniques. One potential direction could involve a deeper investigation into the main factors influencing the thermal efficiency. This could include exploring novel data sources or variables that may have been overlooked in this study. Moreover, considering the influence of external factors such as system design parameters or operating conditions on thermal efficiency could provide valuable insights for refining modeling approaches.

CRedit authorship contribution statement

Badr Elkari: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Taha Ait Tchakoucht:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Writing – original draft, Writing – review & editing. **Tarik Kousksou:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Formal analysis, Conceptualization. **Yassine Chaibi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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References

- Abdelrazik, A.S., Al-Sulaiman, F.A., Saidur, R., Ben-Mansour, R., 2018. A review on recent development for the design and packaging of hybrid photovoltaic/thermal (PV/T) solar systems. *Renew. Sustain. Energy Rev.* 95, 110–129. <https://doi.org/10.1016/j.rser.2018.07.013>.
- Al-Alawi, A., Al-Alawi, S.M., Islam, S.M., 2007. Predictive control of an integrated PV-diesel water and power supply system using an artificial neural network. *Renew. Energy* 32, 1426–1439.
- Alapatt, D., Mascagni, P., Srivastav, V., Padoy, N., 2020. Neural networks and deep learning. *Artif. Intell. Surg.*
- Ali, A., Aurangzeb, K., Shoaib, M., Alhussein, M., Malik, M.Z., 2023. Second-law analysis of nanofluid-based photovoltaic/thermal system modeling and forecasting model based on artificial neural network. *Eng. Anal. Bound. Elem.* 157, 342–352.
- Al-Waeli, A.H.A., Sopian, K., Kazem, H.A., Yousif, J.H., Chaichan, M.T., Ibrahim, A., Mat, S., Ruslan, M.H., 2018. Comparison of prediction methods of PV/T nanofluid and nano-PCM system using a measured dataset and artificial neural network. *Sol. Energy* 162, 378–396. <https://doi.org/10.1016/j.solener.2018.01.026>.
- Al-Waeli, A.H.A., Kazem, H.A., Yousif, J.H., Chaichan, M.T., Sopian, K., 2019. Mathematical and neural network models for predicting the electrical performance of a PV/T system. *Int. J. Energy Res.* 43, 8100–8117. <https://doi.org/10.1002/er.4807>.
- Argiriou, A.A., Bellas-Velidis, I., Kummert, M., André, P., 2004. A neural network controller for hydronic heating systems of solar buildings. *Neural Netw.* 17, 427–440.
- Borup, D., Christensen, B.J., Mühlbach, N.S., Nielsen, M.S., 2023. Targeting predictors in random forest regression. *Int. J. Forecast.* 39, 841–868.
- Chaibi, Y., Allouhi, A., Salhi, M., 2020. A simple iterative method to determine the electrical parameters of photovoltaic cell. *J. Clean. Prod.*, 122363 <https://doi.org/10.1016/j.jclepro.2020.122363>.
- Chaibi, Y., El Rhafiki, T., Simón-Allué, R., Guede, I., Luaces, C., Charro Gajate, O., Kousksou, T., Zeraoui, Y., 2021. Physical models for the design of photovoltaic/thermal collector systems. *Sol. Energy* 226, 134–146. <https://doi.org/10.1016/j.solener.2021.08.048>.
- Chaibi, Y., ElRhafiki, T., Simón-Allué, R., Guede, I., Luaces, S.C., Gajate, O.C., Kousksou, T., Zeraoui, Y., 2021. Air-based hybrid Photovoltaic/Thermal systems: A review Y. *J. Clean. Prod.* 295 <https://doi.org/10.1016/j.jclepro.2021.126211>.
- Chaibi, Y., Malvoni, M., El Rhafiki, T., Kousksou, T., Zeraoui, Y., 2021. Artificial neural-network based model to forecast the electrical and thermal efficiencies of PVT air collector systems. *Clean. Eng. Technol.* 4, 100132 <https://doi.org/10.1016/j.clet.2021.100132>.
- Dalala, Z., Al-Addous, M., Alawneh, F., Class, C.B., 2020. Environmental data set for the design and analysis of the Photovoltaic system in the Jordan Valley. *Data Br.* 31, 105794 <https://doi.org/10.1016/j.dib.2020.105794>.
- De Soto, W., Klein, S.A., Beckman, W.A., 2006. Improvement and validation of a model for photovoltaic array performance. *Sol. Energy* 80, 78–88. <https://doi.org/10.1016/j.solener.2005.06.010>.
- Deka, M.J., Kalita, P., Das, D., Kamble, A.D., Bora, B.J., Sharma, P., Medhi, B.J., 2023. An approach towards building robust neural networks models using multilayer perceptron through experimentation on different photovoltaic thermal systems. *Energy Convers. Manag.* 292, 117395.
- Deka, M.J., Kamble, A.D., Das, D., Sharma, P., Ali, S., Kalita, P., Bora, B.J., Kalita, P., 2024. Enhancing the performance of a photovoltaic thermal system with phase change materials: Predictive modelling and evaluation using neural networks. *Renew. Energy*, 120091.
- Ding, X., Liu, J., Yang, F., Cao, J., 2021. Random radial basis function kernel-based support vector machine. *J. Frankl. Inst.* 358, 10121–10140.
- Evans, D.L., 1981. Simplified method for predicting photovoltaic array output. *Sol. Energy* 27, 555–560.
- Hai, T., Zhou, J., 2023. Predicting the performance of thermal, electrical and overall efficiencies of a nanofluid-based photovoltaic/thermal system using Elman recurrent neural network methodology. *Eng. Anal. Bound. Elem.* 150, 394–399.
- Han, J., Kamber, M., Pei, J., 2012. Data mining concepts and techniques third edition. Univ. Illinois Urbana-Champaign Micheline Kamber Jian Pei Simon Fraser Univ.
- IEA, Electricity Market Report 2023, 2023. doi:(10.1787/f0aed4e6-en).
- Joshi, A.S., Tiwari, A., Tiwari, G.N., Dincer, I., Reddy, B.V., 2009. Performance evaluation of a hybrid photovoltaic thermal (PV/T)(glass-to-glass) system. *Int. J. Therm. Sci.* 48, 154–164.
- Kalani, H., Sardarabadi, M., Passandideh-Fard, M., 2017. Using artificial neural network models and particle swarm optimization for manner prediction of a photovoltaic thermal nanofluid based collector. *Appl. Therm. Eng.* 113, 1170–1177. <https://doi.org/10.1016/j.applthermaleng.2016.11.105>.
- Kalani, H., Sardarabadi, M., Passandideh-Fard, M., 2017. Using artificial neural network models and particle swarm optimization for manner prediction of a photovoltaic thermal nanofluid based collector. *Appl. Therm. Eng.* 113, 1170–1177.
- Karunasingha, D.S.K., 2022. Root mean square error or mean absolute error? Use their ratio as well. *Inf. Sci. (Ny.)* 585, 609–629.
- Krzywinski, M., Altman, N., 2015. Multiple linear regression: when multiple variables are associated with a response, the interpretation of a prediction equation is seldom simple. *Nat. Methods* 12, 1103–1105.
- Mojumder, J.C., Ong, H.C., Chong, W.T., Shamshirband, S., Abdullah-Al-Mamoon, 2016. Application of support vector machine for prediction of electrical and thermal performance in PV/T system. *Energy Build.* 111, 267–277. <https://doi.org/10.1016/j.enbuild.2015.11.043>.
- Mojumder, J.C., Ong, H.C., Chong, W.T., Izadyar, N., Shamshirband, S., 2017. The intelligent forecasting of the performances in PV/T collectors based on soft computing method. *Renew. Sustain. Energy Rev.* 72, 1366–1378. <https://doi.org/10.1016/j.rser.2016.11.225>.
- Montesinos López, O.A., Montesinos López, A., Crossa, J., 2022. Fundamentals of artificial neural networks and deep learning. *Multivar. Stat. Mach. Learn. Methods Genomic Predict.* Springer, pp. 379–425.
- Ravaee, H., Farahat, S., Sarhaddi, F., 2012. Artificial neural network based model of photovoltaic thermal (pv/t) collector. *J. Math. Comput. Sci.* 4, 411–417.
- Sathe, T.M., Dhoble, A.S., 2017. A review on recent advancements in photovoltaic thermal techniques. *Renew. Sustain. Energy Rev.* 76, 645–672. <https://doi.org/10.1016/j.rser.2017.03.075>.
- Shaban, W.M., Kabeel, A.E., Attia, M.E.H., Talaat, F.M., 2024. Optimizing photovoltaic thermal solar systems efficiency through advanced artificial intelligence driven thermal management techniques. *Appl. Therm. Eng.*, 123029.
- Shan, F., Tang, F., Cao, L., Fang, G., 2014. Dynamic characteristics modeling of a hybrid photovoltaic-thermal solar collector with active cooling in buildings. *Energy Build.* 78, 215–221.
- Siecker, J., Kusakana, K., Numbi, B.P., 2017. A review of solar photovoltaic systems cooling technologies. *Renew. Sustain. Energy Rev.* 79, 192–203. <https://doi.org/10.1016/j.rser.2017.05.053>.
- Solanki, S.C., Dubey, S., Tiwari, A., 2009. Indoor simulation and testing of photovoltaic thermal (PV/T) air collectors. *Appl. Energy* 86, 2421–2428.
- Tiwari, S., Agrawal, S., Tiwari, G.N., 2018. PVT air collector integrated greenhouse dryers. *Renew. Sustain. Energy Rev.* 90, 142–159. <https://doi.org/10.1016/j.rser.2018.03.043>.
- Tonui, J.K., Tripanagnostopoulos, Y., 2007. Improved PV/T solar collectors with heat extraction by forced or natural air circulation. *Renew. Energy* 32, 623–637. <https://doi.org/10.1016/j.renene.2006.03.006>.
- Vaishak, S., Bhale, P.V., 2019. Photovoltaic/thermal-solar assisted heat pump system: Current status and future prospects. *Sol. Energy* 189, 268–284. <https://doi.org/10.1016/j.solener.2019.07.051>.
- Venkatesh, K.A., Mishra, D., Manimozhi, T., 2023. Model selection and regularization. *Stat. Model. Mach. Learn. Elsevier*, pp. 159–178.
- Wong, T.-T., 2015. Performance evaluation of classification algorithms by k-fold and leave-one-out cross validation. *Pattern Recognit.* 48, 2839–2846.
- Zamen, M., Baghban, A., Pourkiaei, S.M., Ahmadi, M.H., 2019. Optimization methods using artificial intelligence algorithms to estimate thermal efficiency of PV/T system. *Energy Sci. Eng.* 7, 821–834. <https://doi.org/10.1002/ese3.312>.